LOAN APPROVAL PREDICTION SYSTEM USING MACHINE LEARNING

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING
BACHELOR OF TECHNOLOGY
In
A.Y.: 2026-27

By

Challarpu Ashesh 2520030114

Polisetty Sreeharsha 2520030123

Under the Esteemed guidance of
Dr.G.S.Raghavendra
Associate Professor, CSE

Department of Computer Science and Engineering



Koneru Lakshmaiah Education Foundation
(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

Abstract

The Loan Approval Prediction System is a machine learning project that predicts whether a loan application is likely to be approved or not approved based on applicant details such as income, credit history, employment status, and loan amount. The project includes data analysis, preprocessing, model training, and performance evaluation. Logistic Regression is compared with an existing saved model using metrics such as accuracy, precision, recall, F1-score, and ROC-AUC. The trained model is integrated into an application that accepts applicant details and displays a predicted loan status. The project demonstrates the practical use of machine learning techniques for loan approval prediction.

1. Introduction

1.1 Purpose of the Project

Loan approval involves checking different applicant details such as income, credit history, employment, and loan amount. The purpose of this project is to develop a machine learning system that can learn from previous loan applications and predict whether a new application is likely to be approved or rejected. The project also aims to make the prediction easier to understand by identifying the important factors that influence the model's decision.

1.2 Problem Statement

The loan approval process depends on several applicant-related factors, and evaluating these factors manually can take time. The problem is to develop a **binary classification system** that uses information from a loan application to predict its approval status.

The system will take applicant details as input and produce a predicted loan status as **Approved or Not Approved**. It will also evaluate different machine learning models and analyze the factors that influence their predictions.

The project topic itself is defined as predicting loan approval from information such as applicant income, credit history, employment and loan amount.

1.3 Existing Methods and Disadvantages

Existing loan approval studies use both traditional machine learning models and ensemble methods. Models such as **Logistic Regression, Decision Tree, Random Forest, XGBoost and other boosting methods** have been applied to loan approval and credit-risk prediction. Recent studies also use cross-validation, hyperparameter tuning and explainability techniques such as SHAP and LIME.

However, different models can produce different results depending on the dataset and preprocessing used. Class imbalance can also affect classification performance, making accuracy alone insufficient for evaluating the model. In addition, complex models may be difficult to understand without appropriate explainability methods.

Therefore, our project will focus on comparing multiple models, using suitable evaluation metrics, and providing an understandable explanation of the prediction.

1.4 Proposed System

The proposed system is a machine learning-based loan approval prediction system that predicts whether a loan application is likely to be approved or not approved. The system will use applicant information such as income, credit history, employment status, loan amount, and other relevant details as input. The data will be analyzed and preprocessed before training different machine learning models. The models will be compared using suitable evaluation metrics to identify their performance. The system will also use explainability techniques to identify the

important factors influencing the prediction. Finally, the trained model can be integrated into a simple application where users can enter loan details and receive a prediction. This follows the end-to-end ML workflow covered in the course, from data preparation and model training to evaluation and deployment.

1.5 Research Objectives

1. **To analyze and preprocess loan application data** by handling missing values, encoding categorical features, and preparing the data for machine learning.
2. **To develop and compare different machine learning models** such as Logistic Regression, Decision Tree, Random Forest, and XGBoost for loan approval prediction.
3. **To evaluate the models using suitable performance metrics** such as accuracy, precision, recall, F1-score, and ROC-AUC.
4. **To identify the important factors affecting loan approval predictions** using feature-importance and explainability techniques such as SHAP.
5. **To develop a practical end-to-end ML system** that can be integrated into a simple application for predicting loan approval.

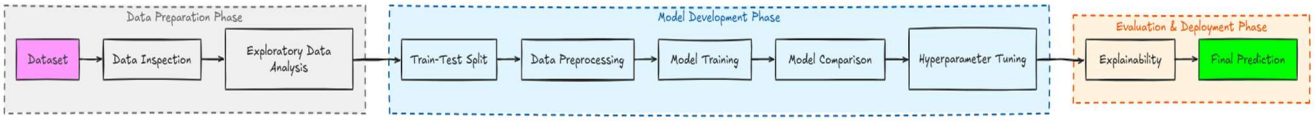
S.No.	Title	Reference with DOI	Problem or gap addressed	Objective of the Research	Focus of the paper (Methods used)	Inference	Recommendations for future research
1	Machine Learning-Based Loan Approval Automation: Enhancing Efficiency, Accuracy and Fairness in Credit Decision-Making	Mani Ghahremani, Hellen Amimo Otieno, Mishanil Kazreen, and Stavros Shiaeles (2026), *Expert Systems*, 43(7), e70307. DOI: **10.1111/exsy.70307**.	Traditional loan approval can be manual and time-consuming. The study also considers class imbalance, model tuning, explainability and fairness.	To develop and compare machine learning models for loan eligibility while improving transparency and fairness.	Random Forest, XGBoost, stacking and voting ensembles; SMOTE; GridSearchCV with 5-fold cross-validation; SHAP and LIME.	The tuned Random Forest achieved strong performance on the study dataset, and SHAP and LIME were used to explain predictions.	The study suggests further work on robustness, fairness and responsible deployment.
2	Advancing Loan Approval Prediction With SHAP-Guided Feature Selection and LIME-Based Model Interpretability in a Multiclassifier Context Through a Web-Based Application Development Approach	Raisa Akter, Rajib Kumar Halder, Mohammed Nasir Uddin, Md. Ashraf Uddin, Ansam Khraisat, Mijanur Rahman, and Md. Kabir Hossain (2026), *International Journal of Intelligent Systems*, 2026(1), 8899164. DOI: **10.1155/int/8899164**	Complex loan-approval models can be difficult to understand, especially when applicant data contains many different types of features.	To improve loan approval prediction using feature selection and explainable machine learning..	Data preprocessing, missing-value handling, categorical encoding, SHAP-guided feature selection, multiple classifiers and LIME-based explanations; a real-time web application was also developed.	The study showed that explainability can be integrated with loan prediction, and the developed web application demonstrates practical use.	The authors identify real-world datasets, fairness-aware learning, drift monitoring and model governance as future directions.
3	A Comparative Performance Assessment for Prediction of Loan Approval in Financial Sector	Lopamudra Hota, Puneet Kumar Jain, and Arun Kumar (2025), *Procedia Computer Science*, 258, 298–307. DOI: **10.1016/j.procs.2025.04.267**.	Different machine learning models can produce different results for loan approval, making	To compare individual classifiers and ensemble methods for loan approval prediction.	Logistic Regression, Decision Tree, Naive Bayes, SVM and Neural Network as baseline models, along with bagging,	The study reports that Logistic Regression performed strongly among the individual models, while	Further evaluation on different datasets and lending environments would help assess robustness.

S.No.	Title	Reference with DOI	Problem or gap addressed	Objective of the Research	Focus of the paper (Methods used)	Inference	Recommendations for future research
			model selection important.		boosting and stacking ensembles.	Random Forest performed best among the ensemble approaches in its experimental setting.	
4	Predictive Modeling for Bank Loan Approval: From Data to Decisions	Davinder Paul Singh, Amit Khare, Purusothaman N., Ruchi Lal, Smriti Singh Chauhan, and Kalaiarasan C. (2025), *Procedia Computer Science*, 259, 1426–1431. DOI: **10.1016/j.procs.2025.04.097**.	Traditional loan approval often depends on manual evaluation, which can be time-consuming and may lead to inconsistent decisions.	To investigate machine learning techniques for automating bank loan approval.	Logistic Regression, Decision Tree, Random Forest, SVM and Gradient Boosting, with evaluation using accuracy, precision, recall and F1-score.	The study demonstrates that machine learning can be used to automate loan approval and systematically compare classification models.	Further work can focus on improving robustness, fairness and practical deployment.
5	An Ensemble Classifier-Based Approach for Feature Selection and Prediction of Loan Approval	Narendra Singh Ranawat (2025), *Indian Journal of Finance*, 19(4), 24–39. DOI: **10.17010/ijf/2025/v19i4/174947**.	Using many input features may increase model complexity, so identifying the most useful features can help build more efficient prediction models.	To use feature ranking and ensemble learning for loan approval prediction.	LightGBM, XGBoost, CatBoost and AdaBoost with grid-search optimisation and feature-sensitivity analysis.	The study found that feature selection combined with ensemble learning could maintain strong performance while reducing the number of features used.	The authors recommend exploring additional sources such as transactional and behavioural data.
6	Loan Approval Classification Using Ensemble Learning on Imbalanced Data	Rahmi Anadra, Kusman Sadik, Agus M. Soleh, and Reka Agustia Astari (2024), *Enthusiastic: International Journal of Applied Statistics and Data Science*, 4(2), 85–95. DOI: **10.20885/enthusiastic.vol4.iss2.art1**.	Loan approval data can contain an imbalance between approved and rejected applications, which can affect classification performance.	To study ensemble learning for loan approval prediction under class imbalance.	Random Forest, XGBoost, SMOTE, data exploration, preprocessing, 10-fold cross-validation and evaluation using accuracy, sensitivity, specificity and F1-score.	The study reported strong performance from XGBoost after handling the class imbalance	Further testing with different datasets and imbalance settings would help assess generalisation.
7	A Comparative Study of Loan Approval Prediction Using Machine Learning Methods	Vahid Sinap (2024), *Gazi Üniversitesi Fen Bilimleri Dergisi Part C: Tasarım ve Teknoloji*, 12(2), 644–663. DOI: **10.29109/gujsc.1455978**	Different machine learning models and feature-selection methods may affect the quality of loan approval prediction.	To compare machine learning algorithms and examine the effect of feature selection.	Logistic Regression, KNN, SVM, Decision Tree and Random Forest, together with K-Best and Recursive Feature Elimination.	The study demonstrates the importance of comparing algorithms and examining feature-selection effects.	Testing additional datasets and model configurations can help evaluate robustness.
8	Predicting Loan Eligibility Approval	Guangxuan Chen (2024), *Proceedings of the 1st International Conference on Data Science and Engineering (ICDSE)*, pp. 512–517.	Financial institutions need data-driven	To apply machine learning algorithms	Machine learning classification approaches	The study demonstrates the use of ML to	Further testing on broader datasets and additional applicant information can

S.No.	Title	Reference with DOI	Problem or gap addressed	Objective of the Research	Focus of the paper (Methods used)	Inference	Recommendations for future research
	Using Machine Learning Algorithms	DOI: **10.5220/0012828200004547**.	approaches to evaluate loan eligibility efficiently.	to loan eligibility prediction.	applied to loan-eligibility data.	support faster and more systematic loan-eligibility assessment.	improve generalisation.

3. Methodology

3.1 Proposed System Flow Diagram



3.2 Dataset Description

The study uses a **loan approval dataset obtained from Kaggle**. The exact CSV used in this project contains **614 records and 13 columns**. The dataset includes applicant-related information such as gender, marital status, number of dependents, education, self-employment status, applicant income, co-applicant income, loan amount, loan amount term, credit history, property area, and loan status.

The target variable is **loan_status**, where **Y represents an approved loan and N represents a loan that is not approved**. The dataset contains **422 approved applications (68.73%) and 192 not-approved applications (31.27%)**. The dataset contains both numerical and categorical attributes and also includes missing values in several columns. These characteristics make the dataset suitable for studying data preprocessing, classification, model comparison, and explainable machine learning.

3.3 Experimental Setup

The project is implemented in **Python using Jupyter notebooks through Visual Studio Code**. The experiments use Python libraries such as **pandas and NumPy** for data handling, **Matplotlib and Seaborn** for exploratory data analysis and visualization, and **scikit-learn** for data preprocessing, machine learning models, model evaluation, and hyperparameter tuning. **XGBoost** is used for gradient-boosting classification and model comparison.

The experiments are carried out in a standard CPU-based development environment. A fixed random state is used wherever applicable to improve reproducibility. Different classification models are trained on the prepared dataset and evaluated using multiple performance metrics rather than relying only on accuracy.

3.4 Data Preprocessing

- The dataset is first inspected to identify the data types, missing values, duplicate records, and the distribution of the target variable. The **loan_id** column is removed because it is an identifier and does not provide useful information for prediction.
- Missing values in categorical features are handled using the **mode**, while missing values in numerical features are handled using the **median**. The categorical variables are converted into numerical form using **one-hot encoding**. The target variable **loan_status** is converted into binary values, where **Y is represented as 1 and N as 0**.

- To prevent data leakage, the dataset is first divided into training and testing data, and preprocessing transformations are fitted only on the training data before being applied to the test data. This ensures that information from the test set is not used during model training. The prepared data is then used for training and evaluating the machine learning models.

3.5 Feature Engineering and Feature Selection

The dataset contains both numerical and categorical features. Categorical features are converted into numerical representations using one-hot encoding, while the identifier column is removed because it does not provide meaningful predictive information.

Feature analysis can help identify applicant attributes associated with loan approval predictions. Model-based feature importance and explainability techniques such as SHAP may be explored to understand the contribution of individual features. Any feature-selection or explainability results will be reported only if they have been implemented and evaluated in the project.

3.6 Model Building and Training

The loan approval problem is treated as a binary classification task, in which the target variable represents approved and not-approved applications. Logistic Regression is evaluated as a classification model, and its performance is compared with that of the existing saved model.

The models are trained using the prepared loan application data. Their predictions are evaluated on held-out test data to measure their ability to classify loan applications. Additional algorithms, such as Decision Tree, Random Forest, and XGBoost, may be evaluated as part of future experimentation if they have not already been implemented.

3.7 Hyperparameter Tuning

Cross-validation helps estimate how consistently a model performs across different subsets of the training data. The project uses cross-validation to compare the performance of the evaluated classification models.

Further hyperparameter tuning can be carried out using GridSearchCV or RandomizedSearchCV to examine different model configurations. Parameters such as the number of trees and maximum tree depth for Random Forest, or the learning rate and number of estimators for XGBoost, can be explored if these models are included in future experiments. Any tuning procedures should be described as completed only when they have actually been performed.

3.8 Model Evaluation

The performance of the trained models is evaluated using multiple classification metrics rather than relying only on accuracy. The main metrics used are **Accuracy, Precision, Recall, F1-score, and ROC-AUC**.

Accuracy measures the overall percentage of correct predictions. Precision measures how many of the applications predicted as approved are actually approved. Recall measures how many of the actual approved applications are correctly identified by the model. F1-score provides a balance between precision and recall. ROC-AUC measures the ability of the model to distinguish between the two target classes.

Confusion matrices and **ROC curves** are also generated to visualize model performance. The results of all tested models are compared, and the selected model will be used for the final prediction system and explainability analysis.

4. Results and Discussion

4.1 Preprocessing Results

The loan approval dataset obtained from Kaggle contains 614 records and 13 columns. It includes applicant details such as gender, marital status, number of dependents, education, employment status, applicant income, co-applicant income, loan amount, loan amount term, credit history, property area, and loan status

During preprocessing, the loan identifier column was removed because it does not provide useful predictive information. Missing values were handled using suitable imputation methods, including median imputation for numerical features and mode imputation for categorical features. Categorical variables were converted into numerical representations to make them suitable for machine learning algorithms.

The target variable, loan status, was used to distinguish between approved and not-approved applications. The dataset was prepared for training and evaluation, and multiple classification models were compared to assess their predictive performance

4.2 Model Performance

Evaluation metric	Random Forest	Logistic Regression
Accuracy	83.87%	84.95%
Precision	82.67%	83.78%
Recall	96.88%	96.88%
F1-score	0.8921	0.8986
ROC-AUC	0.7699	0.8427

The performance of the existing saved model and the Logistic Regression model was evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. The existing saved model achieved an accuracy of 83.87%, while Logistic Regression achieved 84.95%.

Logistic Regression also obtained a slightly higher F1-score of 0.8986 compared with 0.8921 for the existing model. Both models achieved a recall of 96.88%. Logistic Regression recorded a higher ROC-AUC of 0.8427, indicating stronger discrimination between the two classes on this test split.

These results suggest that Logistic Regression is a promising candidate for the loan approval prediction system. However, further cross-validation and testing would be useful before selecting the final model.

4.3 Confusion Matrix Analysis

The confusion matrix helps evaluate how correctly the model classifies loan applications into approved and not-approved categories. It identifies correct predictions as well as false approvals and false rejections.

For the existing saved model, the evaluation produced 16 true rejections, 13 false approvals, 2 false rejections, and 62 true approvals. The Logistic Regression model produced 17 true rejections, 12 false approvals, 2 false rejections, and 62 true approvals.

The results show that both models correctly identified most approved applications in the test dataset. Logistic Regression produced one fewer false approval than the existing saved model while maintaining the same number of false rejections. These results help explain the models' overall performance and highlight the importance of considering both types of classification errors.

4.4 Feature Importance and Prediction Analysis

Loan approval predictions depend on several applicant-related factors, including credit history, income, loan amount, employment status, and property area. Feature analysis can help identify which variables contribute most to the model's predictions.

The project includes analysis of applicant data and comparison of machine learning models to understand loan approval patterns. Further feature-importance analysis can be used to examine the contribution of individual features and improve the interpretability of the prediction system.

The predicted loan status should be treated as a model-generated estimate rather than a guarantee of an actual lending decision.

5. Conclusion

This project developed a machine learning-based Loan Approval Prediction System to predict whether a loan application is likely to be approved or not approved using applicant-related information. The workflow includes data preprocessing, handling missing values, categorical encoding, model training, performance evaluation, and comparison of classification models.

The existing saved model achieved an accuracy of 83.87%, while Logistic Regression achieved 84.95% on the evaluated test split. Logistic Regression also recorded a higher F1-score and ROC-AUC in this comparison, making it a promising candidate for the prediction application.

The project demonstrates how machine learning can support systematic loan approval prediction and how multiple evaluation metrics can be used to compare models. The final system integrates the trained models with an application interface so that users can enter applicant details and receive a predicted loan status. The prediction is intended for educational and analytical purposes and should not replace a complete financial assessment.

6. Future Work

- Improve prediction performance through further hyperparameter tuning and cross-validation.
- Evaluate additional machine learning algorithms and compare their results on the same dataset.
- Apply explainability techniques to understand the factors influencing loan approval predictions.
- Investigate class imbalance and assess performance separately for approved and not-approved applications.
- Test the system on additional and more recent loan application datasets to evaluate generalisation.
- Improve the application interface and validation of user inputs.
- Explore secure deployment and responsible use of the prediction system, including fairness and data privacy considerations.

7. References

[Use one consistent style, preferably APA. Include every cited work and verify DOI and bibliographic information.]

1. Ghahremani, M., Otieno, H. A., Kazreen, M., & Shiaeles, S. (2026). Machine learning-based loan approval automation: Enhancing efficiency, accuracy and fairness in credit decision-making. *Expert Systems, 43*(7), e70307. https://doi.org/10.1111/exsy.70307

2. Akter, R., Halder, R. K., Uddin, M. N., Uddin, M. A., Khraisat, A., Rahman, M., & Hossain, M. K. (2026). Advancing loan approval prediction with SHAP-guided feature selection and LIME-based model interpretability in a multiclassifier context through a web-based application development approach. *International Journal of Intelligent Systems, 2026*(1), Article 8899164. https://doi.org/10.1155/int/2026/8899164

3. Hota, L., Jain, P. K., & Kumar, A. (2025). A comparative performance assessment for prediction of loan approval in financial sector. *Procedia Computer Science, 258*, 298–307. https://doi.org/10.1016/j.procs.2025.04.267

4. Singh, D. P., Khare, A., Purusothaman, N., Lal, R., Singh Chauhan, S., & Kalaiarasan, C. (2025). Predictive modeling for bank loan approval: From data to decisions. Procedia Computer Science, 259, 1426–1431. https://doi.org/10.1016/j.procs.2025.04.097

5. Ranawat, N. S. (2025). An ensemble classifier-based approach for feature selection and prediction of loan approval. *Indian Journal of Finance, 19*(4), 24–39. https://doi.org/10.17010/ijf/2025/v19i4/174947

6. Anadra, R., Sadik, K., Soleh, A. M., & Astari, R. A. (2024). Loan approval classification using ensemble learning on imbalanced data. *Enthusiastic: International Journal of Applied Statistics and Data Science, 4*(2), 85–95. https://doi.org/10.20885/enthusiastic.vol4.iss2.art1
7. Sinap, V. (2024). A comparative study of loan approval prediction using machine learning methods. *Gazi Üniversitesi Fen Bilimleri Dergisi Part C: Tasarım ve Teknoloji, 12*(2), 644–663. https://doi.org/10.29109/gujsc.1455978
8. Chen, G. (2024). Predicting loan eligibility approval using machine learning algorithms. In *Proceedings of the 1st International Conference on Data Science and Engineering (ICDSE)* (pp. 512–517). https://doi.org/10.5220/0012828200004547

Signature of the Guide Course Instructor

Dataset Source: Kaggle — Loan Prediction Dataset.

Dataset Description: The dataset contains 614 loan application records and 13 columns, including applicant information and loan approval status.